Analyzing Transformations with Multiple Parameters

Reading the parameters, evaluating a transformed function, and expanding the transformed rule

Directions

- Every rule on this sheet has the shape $g(x) = a\, f\big(b(x-h)\big) + k$. Work from the inside out: b and h act on x, and a and k act on the output.
- Where a table is given, use it as a picture of the parent function — but write the algebra too.
- Give exact values, not decimals.

1. Let $f(x) = x^2$ and $g(x) = 2f(x-3) + 1$. Find $g(5)$.

Answer: ________________

2. Let $f(x) = x^2$ and $g(x) = f(x+4) - 7$.

(a) Find $g(-2)$.

Answer: ________________

(b) Describe, in order, the transformations that take the graph of $y = f(x)$ to the graph of $y = g(x)$. Name the direction of each shift.

Answer: ________________

3. Let $f(x) = x^2$ and $g(x) = -\dfrac{1}{2}f(x+4) - 3$. Write $g(x)$ in expanded form $ax^2 + bx + c$.

Answer: __________________

4. Let $f(x) = x^2 + 3$. Its values are shown below.

x	-2	-1	0	1	2
$f(x)$	7	4	3	4	7

Let $g(x) = f(2x) - 5$.

(a) Find $g(1)$ using the table.

Answer: __________________

(b) Find $g(2)$. (The table stops before you need it — use the rule for f.)

Answer: __________________

5. Let $f(x) = \sqrt{x}$ and $g(x) = -3f(x) + 2$.

(a) Find $g(9)$.

Answer: ________________

(b) Describe, in order, the transformations that take $y = f(x)$ to $y = g(x)$.

Answer: ________________

6. Let $f(x) = x^2 - 4x$ and $g(x) = f(x-2) + 5$. Write $g(x)$ in expanded form $ax^2 + bx + c$.

Answer: ________________

7. Let $p(x) = x^3$ and $q(x) = -p(x-1) + 2$. Complete the table.

x	0	1	3
$q(x)$		2	

(a) Find $q(0)$.

Answer: ________________

(b) Find $q(3)$.

Answer: ________________

8. Let $f(x) = x^3$ and $g(x) = f(6-x)$.

(a) Find $g(2)$.

Answer: ________________

(b) Rewrite the input in the form $-(x-h)$, then describe, in order, the two transformations that take $y = f(x)$ to $y = g(x)$.

Answer: ________________

9. Let $f(x) = \sqrt{x}$ and $g(x) = 3f(2x - 8) + 1$. Rewrite $g(x)$ in the form $a\sqrt{b(x-h)} + k$, so that every parameter can be read off directly.

Answer: ________________

10. Let $g(x) = 4x^2 + 24x + 41$.

(a) Complete the square to write $g(x)$ in the form $a(x-h)^2 + k$.

Answer: ________________

(b) Describe, in order, the transformations that take $y = x^2$ to $y = g(x)$, and say why the order of the last two matters.

Answer: ________________